# Six math clues · answer key

How Bike Derailleurs Work

Grade 2 math. Each problem regroups ones and tens; addition and subtraction are both included.

### Math 1 · 63

28 + 35 = 63. The 13 ones become 1 ten and 3 ones, making 63.

### Math 2 · 34

72 − 38 = 34. Regroup 72 as 6 tens and 12 ones, then subtract to get 34.

### Math 3 · 65

47 + 18 = 65. Add the regrouped ten to the other tens to make 65 wheel turns.

### Math 4 · 37

61 − 24 = 37. Regroup 61 as 5 tens and 11 ones, leaving 37 cards.

### Math 5 · 84

56 + 28 = 84. Regroup the 14 ones, then add the tens to make 84 minutes.

### Math 6 · 46

93 − 47 = 46. Regroup 93 as 8 tens and 13 ones, so 46 teeth remain.

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Grade 2 math. Six tasks use two-digit addition or subtraction with regrouping and answers no greater than 100.

# Reading & writing · answer key

How Bike Derailleurs Work

### Reading · 1

The shifter pulls or releases the cable; cable pull moves the derailleur one way, and the return spring moves it back the other way.

### Reading · 2

The upper pulley guides the chain toward the selected cog, while the lower pulley and spring take up slack to keep the chain snug.

### Reading · 3

The front chainring has twice as many teeth as the back cog: 32 ÷ 16 = 2, so the wheel turns twice per pedal turn.

### Writing

Accept a spoken, dictated, drawn, or written explanation that links one derailleur part to its job.

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A rear derailleur guides a moving chain between the cassette cogs, while its spring-loaded cage and two pulleys take up slack and keep the chain aligned. A front derailleur uses a cage to move the chain between chainrings. A large back cog is easier for climbing; a small one is harder and gives more wheel turns per pedal turn. For the observation, an adult should place the bike upside down on a flat floor or secure it in a stand, operate the shifter, and supervise every moment. The child may turn only the pedal slowly by hand. Keep all fingers, hair, and loose clothing away from the chain, gears, derailleur, wheel, and spokes whenever anything is turning. Stop all movement before counting teeth, and do not ride for this activity.

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Offer dictation: invite a complete spoken explanation, write it down, and leave room for a picture, label, or added sentence.

# Four shifting clues

How Bike Derailleurs Work

### Quiz 1. What happens after a mechanical shifter pulls its cable?

Cable pull moves the rear derailleur sideways so its guide pulley can line the moving chain up with another cog.

### Quiz 2. What does the lower tension pulley help do?

The lower pulley moves with the spring-loaded cage to take up chain slack and keep the loop snug.

### Quiz 3. How does a front derailleur change gears?

A front derailleur's cage nudges the moving chain sideways from one front chainring to another.

### Quiz 4. Which back cog usually makes a hill easier to pedal?

The biggest back cog is an easier gear: the wheel turns less per pedal turn, so less pedal push is needed.

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